Integration by Substitution

ANSWER KEY

u-substitution

1 Evaluate each integral using substitution:

- **a** $\int 2x(x^2+1)^5\,dx\;\; u=x^2+1:\; =\frac{(x^2+1)^6}{6}+C.$
- **b** $\int \frac{x}{\sqrt{x^2+9}}\,dx\;\; u=x^2+9:\; =\sqrt{x^2+9}+C.$
- **c** $\int \cos(3x)\,dx\;\; u=3x:\; =\frac{1}{3}\sin(3x)+C.$

2 Evaluate $\int x^2e^{x^3}\,dx.$
$u=x^3,\, du=3x^2\,dx:\; =\frac{1}{3}e^{x^3}+C.$

Definite integrals with substitution

3 Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly, changing the limits with $u=x^2+1.$
u goes from 1 to 5: $\frac{1}{2}\int_1^5\sqrt{u}\,du=\frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5=\frac{1}{3}(5\sqrt{5}-1).$

4 Evaluate $\int_0^{\pi/2}\sin^3 x\cos x\,dx.$
$u=\sin x$, limits 0 to 1: $\int_0^1 u^3\,du=\frac{1}{4}.$

5 Evaluate $\int_1^e \frac{\ln x}{x}\,dx.$
$u=\ln x$, limits 0 to 1: $\int_0^1 u\,du=\frac{1}{2}.$

Substitution with trigonometric and exponential integrands

6 Evaluate each integral using substitution:

- **a** $\int \sin^4 x\cos x\,dx\;\; u=\sin x:\; =\frac{\sin^5 x}{5}+C.$
- **b** $\int \frac{e^x}{1+e^x}\,dx\;\; u=1+e^x:\; =\ln|1+e^x|+C.$
- **c** $\int \sec^2 x\tan x\,dx\;\; u=\tan x:\; =\frac{\tan^2 x}{2}+C.$

7 Evaluate $\int \frac{2x+3}{x^2+3x+1}\,dx$, recognizing the numerator as the derivative of the denominator.
$u=x^2+3x+1,\, du=(2x+3)\,dx:\; =\ln|x^2+3x+1|+C.$

Choosing the substitution

8 For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3-1)^7\,dx$ $u=x^3-1$, $du=3x^2dx$: $=\frac{1}{3}\int u^7\,du=\frac{(x^3-1)^8}{24}+C$.

b $\int \tan x\,dx=\int \frac{\sin x}{\cos x}\,dx$ $u=\cos x$, $du=-\sin x\,dx$: $=-\ln|\cos x|+C=\ln|\sec x|+C$.

Substitution with radicals

9 Evaluate $\int x\sqrt{2-x}\,dx$ using the substitution $u=2-x$ (so $x=2-u$).
$u=2-x$, $x=2-u$, $dx=-du$:
$\int(2-u)\sqrt{u}(-du)=\int(u-2)\sqrt{u}\,du=\int(u^{3/2}-2u^{1/2})du=\frac{2}{5}u^{5/2}-\frac{4}{3}u^{3/2}+C=\frac{2}{5}(2-x)^{5/2}-\frac{4}{3}(2-x)^{3/2}+C$
.

10 Evaluate $\int_0^3\frac{x}{\sqrt{x+1}}\,dx$ using $u=x+1$ (so $x=u-1$), changing the limits.
u goes from 1 to 4:
$\int_1^4\frac{u-1}{\sqrt{u}}\,du=\int_1^4(u^{1/2}-u^{-1/2})du=\left[\frac{2}{3}u^{3/2}-2u^{1/2}\right]_1^4=\left(\frac{16}{3}-4\right)-\left(\frac{2}{3}-2\right)=\frac{4}{3}+\frac{4}{3}=\frac{8}{3}$.

Visualizing an area found by substitution

11 The graph shows $f(x)=x\sqrt{x^2+1}$ on $[0,2]$. Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly using $u=x^2+1$, and match it to the shaded area.
u goes from 1 to 5: $\frac{1}{2}\int_1^5\sqrt{u}\,du=\frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5=\frac{1}{3}\left(5\sqrt{5}-1\right)\approx 3.393$, which equals the shaded area under the curve.

Substitution with double angle and reciprocal forms

12 Evaluate $\int\sin x\cos x\,dx$ two ways: (a) using $u=\sin x$, and (b) using the identity $\sin x\cos x=\frac{1}{2}\sin(2x)$. Confirm both give equivalent answers.
(a) $u=\sin x$: $\int u\,du=\frac{\sin^2 x}{2}+C$. (b) $\int\frac{1}{2}\sin(2x)\,dx=-\frac{1}{4}\cos(2x)+C$. Using $\cos(2x)=1-2\sin^2 x$: $-\frac{1}{4}(1-2\sin^2 x)+C=\frac{\sin^2 x}{2}+C'$ — the same family, differing only by the constant.

13 Evaluate $\int\frac{1}{x\ln x}\,dx$ using $u=\ln x$.
$u=\ln x$, $du=\frac{1}{x}dx$: $\int\frac{1}{u}\,du=\ln|u|+C=\ln|\ln x|+C$.

Substitution in motion and rate problems

14 A particle's velocity is $v(t)=t\sin(t^2)$ m/s. Find its displacement from $t=0$ to $t=\sqrt{\pi}$, using $u=t^2$.
$u=t^2$, $du=2t\,dt$, limits 0 to π: $\frac{1}{2}\int_0^\pi\sin u\,du=\frac{1}{2}[-\cos u]_0^\pi=\frac{1}{2}(1+1)=1$ m.

15 Evaluate $\int\frac{(\ln x)^2}{x}\,dx$ using $u=\ln x$.
$u=\ln x$, $du=\frac{1}{x}dx$: $\int u^2\,du=\frac{u^3}{3}+C=\frac{(\ln x)^3}{3}+C$.

Recognizing when substitution simplifies a problem

16 Evaluate $\int (x+1)\sqrt{x^2+2x+5}\,dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.

$u = x^2 + 2x + 5$, $du = (2x+2)dx = 2(x+1)dx$: $\frac{1}{2}\int \sqrt{u}\,du = \frac{1}{2} \cdot \frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2 + 2x + 5)^{3/2} + C$.